

SQL Interview Questions

Questions Only

1. SQL Fundamentals

1. What is SQL?
2. What is a database?
3. What is a DBMS?
4. What is an RDBMS?
5. What is a table?
6. What is a row?
7. What is a column?
8. What is a schema?
9. What is a primary key?
10. What is a foreign key?
11. What is a candidate key?
12. What is a composite key?
13. What is a unique key?
14. What is a constraint?

2. SQL Commands

15. What is DDL?
16. What is DML?
17. What is DQL?
18. What is DCL?
19. What is TCL?
20. What is the difference between DELETE, DROP, and TRUNCATE?
21. What is the difference between WHERE and HAVING?
22. What is the purpose of ORDER BY?
23. What is DISTINCT?
24. What is the purpose of LIMIT?

3. SQL Joins

25. What is a JOIN?

- 26. What is an INNER JOIN?
- 27. What is a LEFT JOIN?
- 28. What is a RIGHT JOIN?
- 29. What is a FULL OUTER JOIN?
- 30. What is a CROSS JOIN?
- 31. What is a SELF JOIN?
- 32. What is the difference between INNER JOIN and LEFT JOIN?
- 33. How do you join three or more tables?
- 34. What happens when a join condition is missing?

4. Aggregation and Grouping

- 35. What is GROUP BY?
- 36. What is HAVING?
- 37. What are aggregate functions?
- 38. What are COUNT, SUM, AVG, MIN, and MAX?
- 39. What is the difference between COUNT(*) and COUNT(column)?
- 40. Can you use WHERE with aggregate functions?
- 41. Can you use HAVING without GROUP BY?
- 42. How do you find the total salary by department?
- 43. How do you find departments with more than five employees?

5. Subqueries and CTEs

- 44. What is a subquery?
- 45. What is a correlated subquery?
- 46. What is the difference between a subquery and a join?
- 47. What is a CTE?
- 48. Why are CTEs useful?
- 49. What is a recursive CTE?
- 50. What is the difference between a CTE and a temporary table?

6. Window Functions

- 51. What is a window function?
- 52. What is the difference between GROUP BY and window functions?

- 53. What is ROW_NUMBER()?
- 54. What is RANK()?
- 55. What is DENSE_RANK()?
- 56. What is the difference between RANK and DENSE_RANK?
- 57. What are LAG() and LEAD()?
- 58. What is PARTITION BY?
- 59. How would you find the second-highest salary using a window function?
- 60. How would you find the top three employees in each department?

7. SQL Interview Problems

- 61. How do you find the second-highest salary?
- 62. How do you find the nth-highest salary?
- 63. How do you find duplicate records?
- 64. How do you remove duplicate records?
- 65. How do you find employees who do not have a department?
- 66. How do you find employees earning more than their manager?
- 67. How do you find the highest salary in each department?
- 68. How do you find the top three salaries in each department?
- 69. How do you find customers who have never placed an order?
- 70. How do you find the latest order for each customer?
- 71. How do you calculate a running total?
- 72. How do you calculate month-over-month growth?

8. Database Design and Performance

- 73. What is normalization?
- 74. What are 1NF, 2NF, and 3NF?
- 75. What is denormalization?
- 76. What is an index?
- 77. Why are indexes used?
- 78. What are the disadvantages of indexes?
- 79. What is a clustered index?
- 80. What is a non-clustered index?
- 81. What is a composite index?

- 82. What is a query execution plan?
- 83. What is a transaction?
- 84. What are ACID properties?
- 85. What is a deadlock?
- 86. What is database partitioning?

9. Advanced SQL

- 87. What is a view?
- 88. What is a materialized view?
- 89. What is a stored procedure?
- 90. What is a function in SQL?
- 91. What is a trigger?
- 92. What is a cursor?
- 93. What is dynamic SQL?
- 94. What is NULL in SQL?
- 95. How does NULL behave in comparisons?
- 96. What is COALESCE()?
- 97. What is CASE WHEN?
- 98. What is a recursive query?